

The shared concrete mechanism is:

multiplicative scale  $\longrightarrow$  additive depth,  
 fine metric detail  $\longrightarrow$  coarse order or valuation data,  
 ordinary geometry  $\longrightarrow$  tree, fan, quotient skeleton.

For tropicalization with  $0 < b < 1$ , the meaningful expression is not a fixed  $\log_b(x)$  as  $b \rightarrow 0$ , since fixed  $x$  mostly degenerates. The meaningful move is to write

$$x(b) = b^a,$$

so that

$$\log_b x(b) = a.$$

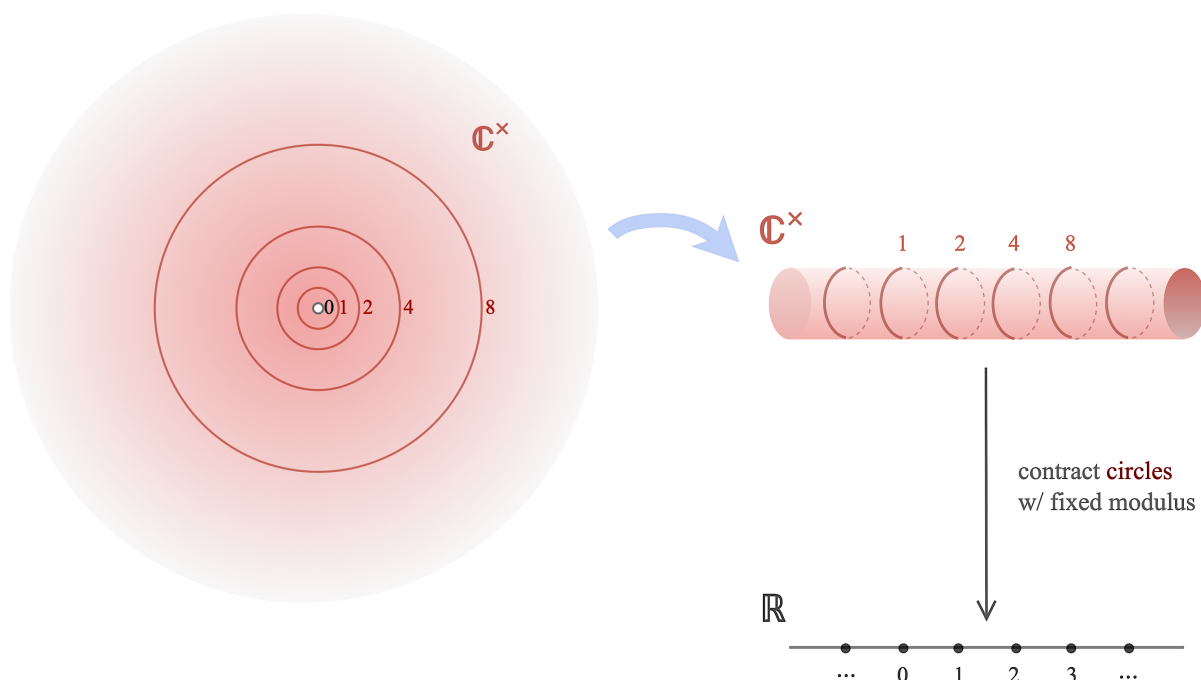


Figure 1: [...]

For sums, one obtains

$$\log_b(b^a + b^c) \longrightarrow \min(a, c) \quad \text{as } b \rightarrow 0.$$

Thus tropicalization says:

keep the exponent or scale;  
 throw away coefficient-level detail;  
 turn addition into min;  
 turn multiplication into addition.

Now compare the binary contraction tower. Take a dyadic grid. At depth  $k$ , cells have size

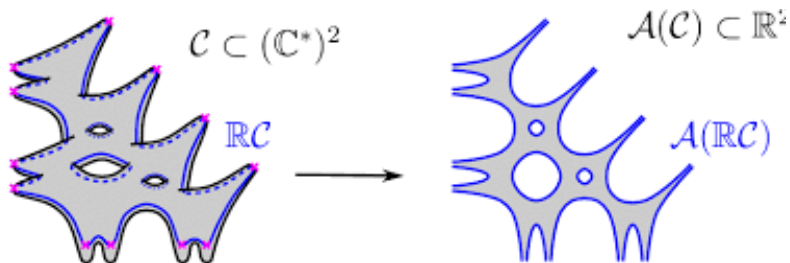


Figure 2: From a paper by Lionel Lang: Applying  $\log | - |$  to curves in a toric variety tend to produce tropical-looking things that aren't quite polyhedral, and look sort of like biological amoebas posing as polyhedral geometry.

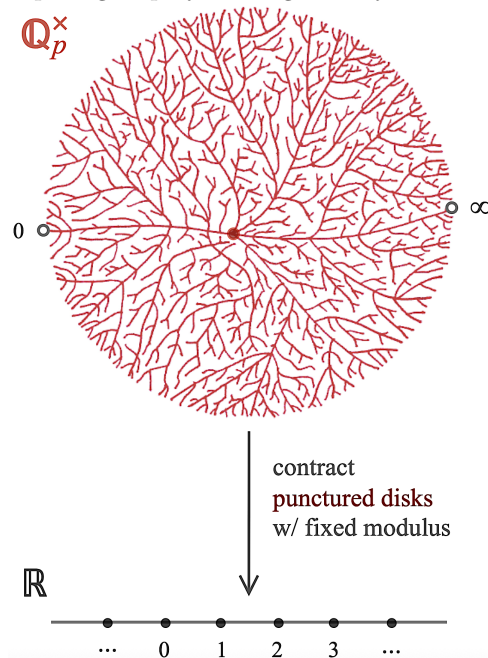


Figure 3: [...]

$$2^{-k}.$$

A binary quotient or coset tier forgets all detail below that scale. A point is no longer “this exact grid point”; it is “this dyadic coset at depth  $k$ .” The important invariant becomes

$$\text{scale} = 2^{-k},$$

and therefore

$$\text{depth} = -\log_2(\text{scale}) = k.$$

So the binary tower is also doing:

keep the exponent or depth;  
throw away within-cell detail.

That is the same move as tropicalization.

The deep reason these feel like the same phenomenon is that both are valuation machines. A valuation converts size or order into an additive number:

$$v(xy) = v(x) + v(y),$$

and

$$v(x + y) \geq \min(v(x), v(y)).$$

In the tropical limit, this inequality becomes the governing rule:

$$v(x + y) = \min(v(x), v(y))$$

generically.

In the binary coset tower, the valuation is dyadic depth:

$$v_2(\text{scale}) = \text{depth}.$$

Equivalently, for two grid points,

$$\text{distance scale} \sim 2^{-k} \quad \Longleftrightarrow \quad \text{separation depth} \sim k.$$

Two points are indistinguishable until the tier where their binary addresses split. The tower keeps that split-depth and discards the lower bits. That is a tropical or ultrametric kind of geometry. So a useful explanation is:

Tropicalization and binary contraction towers are both ways of replacing metric magnitude by scale depth. The logarithm is not an extra analytic trick; it is the coordinate change from multiplicative refinement to additive tier index.

Or, more compactly:

$$\text{tropicalization:} \quad x = b^a \longmapsto a,$$

$$\text{binary tower:} \quad \text{cell size} = 2^{-k} \longmapsto k.$$

Both say: the real object is not the tiny quantity itself, but the exponent at which it lives.

That is why  $\lim_{b \rightarrow 0} \log_b$  and a tower of binary cosets can be understood as manifestations of the same idea: each builds a limiting skeleton where only dominance, scale, and ancestry survive.